

Predicting Diabetes Using Logistic Regression with Pima Indians Diabetes Database

Neelov Alam

Abstract

This paper describes a proposed predictive model intended to evaluate the probability a patient can be diagnosed with diabetes. The statistical theorem used to create the model is logistic regression. The data it uses to generate predictions originates from the "Pima Indians Diabetes Database," [1] sourced from the National Institute of Diabetes and Digestive and Kidney Diseases. The data includes predictive measurements and outcome from a group of 1036 Pima Indian females aged 21 or more. The measurements recorded are as follows: number of pregnancies, glucose levels, blood pressure (mm Hg), skin thickness (mm), insulin levels (mu U/ml), BMI (weight in kg/(height in m)²), diabetes pedigree function, and age (years). This paper's primary objective is to showcase a model that is capable of predicting if a patient has diabetes.

The program is coded in Java using both Java and JavaFX libraries. It is programmed in object-oriented style with six different classes and various helper methods. It has a running time of $O(n)$. The program outputs the average of 8 different probabilities inserted into sigmoid functions with coefficients calculated through logistic regression methods. The program has a 93% success rate with a threshold of 50% and 20% success rate with a high scrutiny threshold of 90%. This paper evaluates the need for different machine learning methods paired with logistic regression in future studies to ensure better accuracy in predicting diabetes diagnoses. It also recommends the use of datasets with more diverse demographics as the results of this paper is limited to females aged 21 or more in the Pima Indian Community.

1. Introduction

In a room of ten Americans, around three or four of them would have prediabetes and 80% of them would not even know it. On top of this, two of them would have type 1 or 2 diabetes and a 20% of them would not even know it. In the United States, diabetes is a concerningly widespread disease, and it only seems to be getting worse, with the number of Americans diagnosed with diabetes having more than doubled in the last 20 years [2].

What makes this chronic disease particularly worrying is that it is often undiagnosed in patients. Type 2 diabetes, the most common variant of the disease, demonstrates little noticeable signs after onset [3]. Moreover, the disease can only be reliably diagnosed with a blood glucose test, such as an FPG or A1C test [4]. This means that numerous patients do not receive proper diagnostic evaluation until symptoms have significantly worsened. Based on this, it is clear that the medical field requires a new method to predict if someone has diabetes without relying on outwardly exhibited symptoms.

Some methods to address this issue have been to develop software programs that use machine-learning models to predict whether a patient has diabetes. Notable studies in the past have used CATBoost and Random Forest to create such programs[6]. In this paper, I use data from The National Institute of Diabetes and Digestive and Kidney Diseases's *Pima Indians Diabetes Database* [1] to develop a similar program. However, in contrast to past studies, I use binary logistic regression to determine the likelihood that a patient has diabetes. This intends to aid medical professionals in assessing how urgently a patient should be evaluated with a blood glucose test before symptoms worsen.

1.1: Rationale in using Logistic Regression:

The choice to use logistic regression as the primary machine learning model in the program is motivated by several factors. The first is that it is efficient in determining probabilities based on numerous categorical variables. In addition, it is suitable for binary classification, such as the presence of evidence indicating diabetes or lack of such evidence. In this regard, it is superior to other modeling methods such as linear regression which is limited in its input of variables and is not suited for binary outputs. In addition, compared to other machine learning models, logistic regression is easier to train in AI applications and grants machine learning programmers a notable degree of flexibility.

1.2 Significance of the Study:

As mentioned, this paper continues research on machine learning modeling techniques used in programs to predict the presence of diabetes. This paper contributes to the existing literature by adding logistic regression as a primary modeling technique whereas other studies have used other techniques. This diversifies the techniques used in the study of predicting diabetes with statistical models.

For medical applications, this paper intends to aid medical health professionals in diagnosing patients with diabetes. Additionally, this paper intends to help those displaying signs of diabetes to acquire treatment as soon as possible. It must be noted; however, that the intention of this paper is *not* to produce a program that can diagnose diabetes. Currently, only a blood glucose test can accurately do this. The intention of this paper is to produce a program modeled after logistic regression that outputs the probability a patient *may* have diabetes.

In terms of financial impact, this paper intends to save medical establishments's expenses by limiting the amount of false-alarm blood glucose tests administered. If the program outputs a low probability that a patient has diabetes, a medical professional may be more reluctant to administer a test, saving the establishment test-related costs. Again, it must be noted that this program does not intend to replace blood glucose tests as a means of determining a diagnosis. It merely intends to serve as an assistive tool. This paper urges medical professionals to administer a test if they believe it appropriate regardless of the output from this program.

1.3 Objectives:

The objectives of this paper are as follows:

1. Describe any related work
2. Demonstrate the methodology of the final software program, which includes any machine-learning methods used, the dataset used, and diagrams demonstrating use of the program
3. Demonstrate results of implementing program
4. Discuss any complications and areas of improvement, as well as recommendations for future studies

2. Related Work

A study in 2022 [6] created an application derived from a Bayesian network (BN) classifier to predict the probability that a user may have diabetes. It used a tree-augmented BN with 11 variables derived from another study to be used in the program. While this study does

not use a BN classifier for its interface, it provides pivotal information to draw from in the creation of this paper.

3. Proposed Methodology

In this methodology, the paper demonstrates the dataset used, the machine learning methods used in the program and its respective mathematical theory, the UML diagram of the application, the class diagram of the program, an overview the overall coding structure.

3.1: Dataset

Table 1 shows the data from *Pima Indians Diabetes Database* [1] used as risk factors in this study.

Pregnan cies	Glucose	BloodPre ssure	SkinThic kness	Insulin	BMI	DiabetesPedigree Function	Age	Outcome
6	148	72	35	0	33.6	0.627	50	1
1	85	66	29	0	26.6	0.351	31	0
8	183	64	0	0	23.3	0.672	32	1
1	89	66	23	94	28.1	0.167	21	0
0	137	40	35	168	43.1	2.288	33	1
5	116	74	0	0	25.6	0.201	30	0
3	78	50	32	88	31	0.248	26	1
10	115	0	0	0	35.3	0.134	29	0
2	197	70	45	543	30.5	0.158	53	1
8	125	96	0	0	0	0.232	54	1
4	110	92	0	0	37.6	0.191	30	0
10	168	74	0	0	38	0.537	34	1
10	139	80	0	0	27.1	1.441	57	0
1	189	60	23	846	30.1	0.398	59	1
5	166	72	19	175	25.8	0.587	51	1
7	100	0	0	0	30	0.484	32	1
Click here to view rest of data								

Table 1: Selected Portion of Data from *Pima Indians Diabetes Database* [1]

3.2: Methods

This section of the paper will describe any machine-learning methods used in the process of creating the program as well as its respective mathematical theory.

3.2.1: Binary Logistic Regression (BLR)

The primary mathematical model used in this study is BLR. There are eight different implementations in the program that use BLR, one for each risk factor. The user will enter variables and the program will output the average of the probabilities (from 0 to 1) determined by the BLR function.

There are 4 main phases in the implementation of BLR:

1. Phase 1: The BLR initializes two coefficients (Θ_1 and Θ_2) for the sigmoid function.
2. Phase 2: The BLR determines the log-likelihood function
3. Phase 3: The BLR uses Newton-raphson method to determine the maximum likelihood estimate
4. Phase 4: Using the sigmoid function with the updated coefficients, the BLR outputs a probability between 0 and 1 based on the user-entered risk factor.

3.2.2: Logit

The program will also use logit for every data point after it is projected onto the log-likelihood function.

The probability will be stored in variable p . It's relationship to $e^{\log(odds)}$ is shown below [7]:

$$p = \frac{e^{\log(odds)}}{1 + e^{\log(odds)}}$$

This relationship is determined by exponentiating both sides:

$$\begin{aligned} \left(\frac{p}{1-p} \right) &= e^{\log(odds)} \\ p &= (1-p) * e^{\log(odds)} \\ p &= (e^{\log(odds)} - p * e^{\log(odds)}) \\ p + p * e^{\log(odds)} &= e^{\log(odds)} \\ p(1 + e^{\log(odds)}) &= e^{\log(odds)} \\ p &= \frac{e^{\log(odds)}}{1 + e^{\log(odds)}} \end{aligned}$$

3.2.2: Sigmoid-function

The program outputs probability based on the sigmoid function.

$$h(x) = 1 / (1 + e^{-z}), \quad z = \theta_1 x + \theta_2$$

3.2.3: Log-likelihood function

The program creates a log-likelihood function to provide the coefficients which will later be used to determine the maximum likelihood estimate.

The following mathematical structures show the the log-likelihood function and the procedures involved in determining it:

$$\text{for samples labelled as 1 : } \prod_{s \text{ in } y_i = 1} p(x_i)$$

$$\text{for samples labelled as 0 : } \prod_{s \text{ in } y_i = 0} (1 - p(x_i))$$

$$L(\beta) = \prod_{s \text{ in } y_i = 1} p(x_i) * \prod_{s \text{ in } y_i = 0} (1 - p(x_i))$$

$$l(\beta) = \sum_{i=1}^n y_i \log \left(\frac{1}{1 + e^{-\beta x_i}} \right) + (1 - y_i) \log \left(\frac{e^{-\beta x_i}}{1 + e^{-\beta x_i}} \right)$$

$$l(\beta) = \sum_{i=1}^n y_i \left[\log \left(\frac{1}{1 + e^{-\beta x_i}} \right) - \log \left(\frac{e^{-\beta x_i}}{1 + e^{-\beta x_i}} \right) \right] + \log \left(\frac{e^{-\beta x_i}}{1 + e^{-\beta x_i}} \right)$$

$$l(\beta) = \sum_{i=1}^n y_i [\log (e^{\beta x_i})] + \log \left(\frac{e^{-\beta x_i}}{1 + e^{-\beta x_i}} * \frac{e^{\beta x_i}}{e^{\beta x_i}} \right)$$

$$l(\beta) = \sum_{i=1}^n y_i \beta x_i + \log \left(\frac{1}{1 + e^{\beta x_i}} \right)$$

$$l(\beta) = \sum_{i=1}^n y_i \beta x_i - \log (1 + e^{\beta x_i})$$

3.2.4: Newton-raphson method

The program will use the Newton-raphson method to determine the coefficients corresponding to the maximum likelihood estimate.

$$\beta^{t+1} = \beta^t - \frac{\nabla_{\beta} l(\beta^t)}{\nabla_{\beta\beta} l(\beta^t)}$$

where,

$\nabla_{\beta} l(\beta^t)$ is first order derivative (gradient)

$\nabla_{\beta\beta} l(\beta^t)$ is second order derivative

t is current iteration number

$$\nabla_{\beta} l = \nabla_{\beta} \sum_{i=1}^n y_i \beta x_i - \log(1 + e^{\beta x_i})$$

$$\nabla_{\beta} l = \sum_{i=1}^n \nabla_{\beta} [y_i \beta x_i - \log(1 + e^{\beta x_i})]$$

$$\nabla_{\beta} l = \sum_{i=1}^n \nabla_{\beta} [y_i \beta x_i] - \nabla_{\beta} [\log(1 + e^{\beta x_i})]$$

$$\nabla_{\beta} l = \sum_{i=1}^n y_i x_i - \left[\frac{1}{(1 + e^{\beta x_i})} * e^{\beta x_i} x_i \right]$$

$$\nabla_{\beta} l = \sum_{i=1}^n y_i x_i - \left[\frac{1}{(1 + e^{-\beta x_i})} * x_i \right]$$

The gradient matrix is determined in the following steps

$$\nabla_{\beta} l = \sum_{i=1}^n y_i x_i - [p(x_i) * x_i]$$

$$\nabla_{\beta} l = \sum_{i=1}^n [y_i - p(x_i)] * x_i$$

$$\nabla_{\beta} l = X^T(Y - \hat{Y})$$

The hessian matrix

$$\nabla_{\beta\beta}l = \nabla_{\beta} \sum_{i=1}^n [y_i - p(x_i)] * x_i$$

$$\nabla_{\beta\beta}l = \sum_{i=1}^n \nabla_{\beta} [y_i - p(x_i)] * x_i$$

$$\nabla_{\beta\beta}l = \sum_{i=1}^n \nabla_{\beta} p(x_i) * x_i$$

$$\nabla_{\beta\beta}l = \sum_{i=1}^n \nabla_{\beta} \left[\frac{1}{1 + e^{-\beta x_i}} \right] * x_i$$

$$\nabla_{\beta\beta}l = \sum_{i=1}^n \left[\frac{1}{1 + e^{-\beta x_i}} \right]^2 e^{-\beta x_i} (-x_i) x_i$$

$$\nabla_{\beta\beta}l = \sum_{i=1}^n \left[\frac{e^{-\beta x_i}}{1 + e^{-\beta x_i}} \right] \left[\frac{1}{1 + e^{-\beta x_i}} \right] x_i^T x_i$$

$$\nabla_{\beta\beta}l = \sum_{i=1}^n p(x_i)(1 - p(x_i))x_i^T x_i$$

$$\nabla_{\beta\beta}l = -X^T P(1 - P)X$$

$$\nabla_{\beta\beta}l = -X^T W X$$

The overall matrix representation of the Newton-raphson method:

$$\begin{aligned}\beta^{(t+1)} &= \beta^{(t)} - \frac{\nabla_{\beta} l(\beta^t)}{\nabla_{\beta\beta} l(\beta^t)} \\ \beta^{(t+1)} &= \beta^{(t)} - \frac{X(Y - \hat{Y}^{(t)})}{(-X^T W^{(t)} X)} \\ \beta^{(t+1)} &= \beta^{(t)} + X(Y - \hat{Y}^{(t)})(X^T W^{(t)} X)^{-1}\end{aligned}$$

3.5: Code Overview

3.5.1: Language and Supplemental Programs

The language used for this program is JavaFX . The program uses FXML files to develop the user interface. It also uses the program SceneBuilder to develop the user interface FXML. Additionally, data used in this program is imported from a CSV file.

3.5.2: Organization

The program utilizes object-oriented programming in standard Java semantic and syntax. It contains five total classes and four supplementary classes:

- Main.java
- parse.java
- subject.java
- vector.java
- logRegression.java
- mainSceneController.java

3.5.3: Code Flow and Overview

- Main.java: Main method is called and scene is built
- Main.java: Main method is called and parse.parseFile() is called
- parse.java: parseFile() reads diabetes.csv (file storing all data used in this study) and initializes array sbjs of size 768 (amount of lines in file) and then iteratively inserts each element in a line into array stor_arr of size 9
- parse.java: parseFile() iteratively stores each data point into each subject object in sbjs

- subject.java: Responsible for providing an object that stores data points, sbjs is a subject array initialized in parse.java to store every data point (9 elements) in each line (768 elements) of data
- parse.java: ParseFile() calls vector.createVectors()
- vector.java: createVectors() calls makeVector() for each type of data (9 times) and stores all factor information in one vector, then stores all of this into one main Vector called list, then it calls logRegression.createData()
- logRegression.java: createData calls newtons_method to return the required coefficient used in the regression equation
- logRegression.java: Once all computations are complete, allCalcsCompleted is set to true, and user interface is enabled
- mainSceneController.java: User inputs all TextFields representing risk factors otherwise an error is thrown
- mainSceneController.java: All TextFields are converted into double and entered into logRegression.sigmoid()
- logRegression.java: sigmoid() returns probability for each risk factor and mainSceneController.show() is called
- mainSceneController.java: show() calculates the mean and standard deviation of all probabilities and shows the user the final probability

3.4: UML Diagram

Figure 1 illustrates the user's path as he or she uses the program.

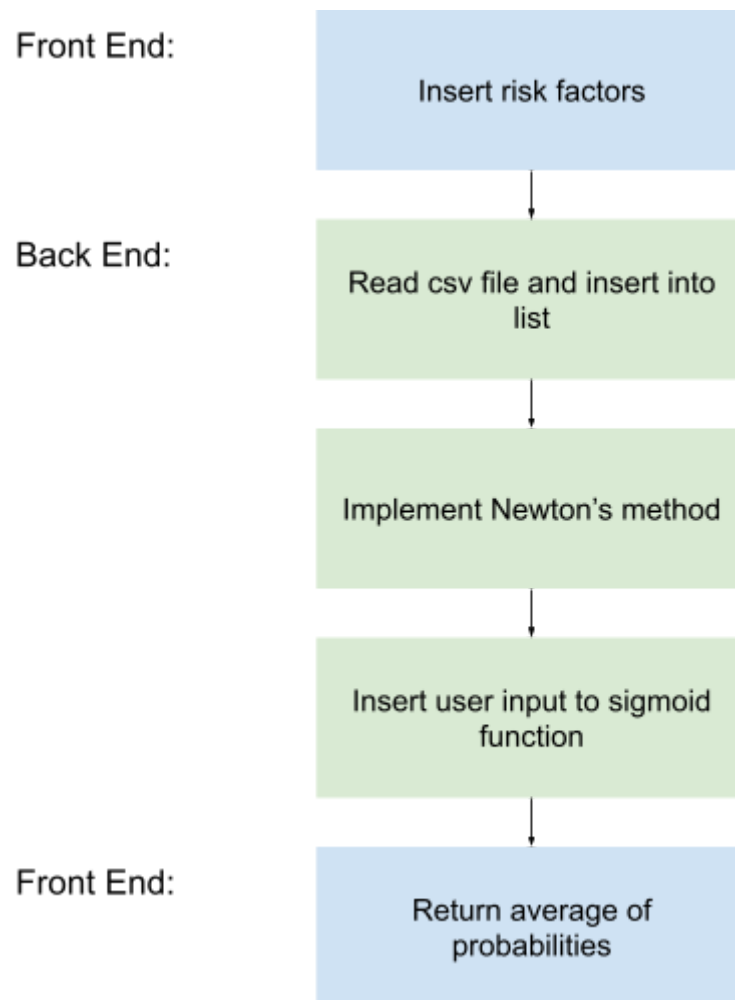


Figure 1: UML diagram of program

3.3: Class Diagram:

Figure 2 illustrates the class diagram of the program.



Figure 2: Class Diagram

4. Experimental Setups and Result Discussion

4.1: Program Validity and Usability Experiment Setup

This experiment is designed to test the accuracy of the program in determining whether a patient has diabetes.

The setup of the experiment is as follows:

- There are 3 trials in which 5 random, independently selected participants in the *Pima Indians Diabetes Database* will be selected.
- If there are any repeat participants (say in both trial 1 and 2 participant 50 was selected), the random sampling process will be repeated until there are no repeat participants.
- These 5 participant's risk factors will be entered into the program
- The paper will note the output of the program for each participant and compare it to the outcome of diabetes testing (0 or 1) of each participant.
- If the output is greater than 0.50 is the actual outcome for the participant is 1, or it is less than 0.50 if the actual outcome is 0, this will be considered a success.
- For further scrutiny, the experiment will take note of whether the output of the program is greater than or equal to 0.90 for an outcome of 1 or less than or equal to 0.10 for an outcome of 0.

4.2: Results

The following tables showcase the results:

Trial 1:

Participant	Preg.	Glucose	B.P.	S.T.	Insulin	BMI	D.P.F.	Age	Outcome	Success	S. Test
579	10	133	68	0	0	27	0.245	36	0	Yes	No
281	0	146	70	0	0	37.9	0.334	28	1	Yes	No
266	5	96	74	18	67	33.6	0.997	43	0	Yes	No
748	1	81	74	41	57	46.3	1.096	32	0	Yes	Yes
512	0	139	62	17	210	22.1	0.207	21	0	Yes	No

Trial 2:

Participant	Preg.	Glucose	B.P.	S.T.	Insulin	BMI	D.P.F.	Age	Outcome	Success	S. Test
77	7	62	78	0	0	32.6	0.391	41	0	Yes	No

668	1	107	50	19	0	28.3	0.181	29	0	Yes	Yes
369	3	81	86	16	66	27.5	0.306	22	0	Yes	No
623	6	183	94	0	0	40.8	1.461	45	0	Yes	No
8	10	115	0	0	0	35.3	0.134	29	0	Yes	No

Trial 3:

Participant	Preg.	Glucose	B.P.	S.T.	Insulin	BMI	D.P.F.	Age	Outcome	Success	S. Test
492	2	89	90	30	0	33.5	0.292	42	0	Yes	No
702	6	125	78	31	0	27.6	0.565	49	1	Yes	No
82	2	74	0	0	0	0	0.102	22	0	Yes	Yes
750	6	162	62	0	0	24.3	0.178	50	0	No	No
462	9	120	72	22	56	20.8	0.733	48	0	Yes	No

4.2: Results Discussion

14 of 15 total experimental trials demonstrated success in predicting diabetes (output of less than or greater than 0.50 depending on outcome). However, only 3 of 15 trials passed the scrutiny test (output of greater than or equal to 0.90 or less than or equal to 0.10 depending on the outcome). It must be acknowledged that in one of these successes, one had incomplete or invalid testing results (B.P., S.T., and insulin in participant 82) which may have influenced the output of the program.

It must also be acknowledged that the program outputs were tested against participants whose information was used in the development of the outputs itself. However, there are 768 total participants in the dataset. This equates to the influence of each participant on the program output at $1/768$ or around 0.130%. Given this relatively negligible level of influence, it is safe to assume that the use of each participant's risk factor did not skew the results of the experiment.

Regarding the results of the experiment, the program demonstrates exceptional ability to predict diabetes with a threshold of 50%. However, as the results of the scrutiny tests indicate, the program's ability to predict diabetes worsens the higher the threshold is set. This indicates

that the program is a fair estimator of the presence of diabetes in patients. However, it is not flawless in its estimation, particularly if the level of scrutiny is high.

5. Comparison

A study published in the National Library of Medicine created a similar computer program using decision tree, SVM, Random Forest, Logistic Regression, KNN, and various other similar techniques to output (in the XGBoost classifier using the ADASYN approach) of 81% [8]. This paper demonstrates a success rate of 93% with a set threshold of 50% and 20% success rate with a high-scrutiny threshold of 60%.

6. Conclusion

The objective of this paper was to demonstrate the development of a computer program that uses logistic regression to predict the presence of diabetes in patients showing certain risk factors. The program distinguishes itself from similar programs in that it uses exclusively logistic regression in an academic setting where it has not been to a large extent.

The program used JavaFX methods in object-oriented style programming. It includes six different classes with one main class and five inherited classes.

The program demonstrated a success rate of 0.9333 with a threshold of 50% and 0.2000 with a threshold of 90%. While its ability to predict diabetes with a marginal threshold of 50% is exceptional, it starts to show weakness when the threshold is set higher. Particularly, in high-scrutiny situations where the threshold is set to 90%, it shows a success rate of only 0.2000. This indicates that further research is required to account for outliers and incomplete or invalid data entries to ensure a higher success rate with a high scrutiny threshold.

Additionally, further research should use databases from various diverse demographics, as the data used in this study was isolated to females aged 21 or more in the Pima Indian Community. Results from different demographics allow for a larger window of applicability for various types of users. Additionally, other machine-learning methods should be used separately or in addition to logistic regression to account for variability in different datasets.

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Author contributions

This paper was authored by one individual who contributed throughout.

Ethical standard

This article does not contain any studies with human participants or animals performed by the author.

Data availability

All data used in development of the program is available on <https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database/>.

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